

While Loops In Java

Generated Aug 13, 2026 7:57 AM

What We'll Learn

By the end of this lesson, students can:

- Explain how a while loop repeats statements.
- Identify the loop condition and loop body.
- Create a counter-controlled while loop.
- Use a while loop with user input.
- Prevent common problems such as infinite loops.

Before We Start

Imagine that you need to display the same message five times.

You could write:

```
System.out.println("Welcome to Java!");  
System.out.println("Welcome to Java!");  
System.out.println("Welcome to Java!");  
System.out.println("Welcome to Java!");  
System.out.println("Welcome to Java!");
```

This works, but repeating the same statement makes the program longer and more difficult to update.

How can we tell Java to repeat a statement without writing it several times?

Let's Understand

A loop allows a program to repeat a group of statements.

A while loop repeatedly executes its statements while a condition remains true.

Basic Syntax

```
while (condition) {  
    // Statements to repeat  
}
```

The condition is checked before the statements inside the loop are executed.

How A While Loop Works

1. Java checks the condition.
2. If the condition is true, the loop body executes.
3. The program returns to the condition.
4. The condition is checked again.
5. The loop continues while the condition is true.
6. The loop stops when the condition becomes false.

Creating A Counter-Controlled Loop

A counter-controlled loop uses a variable to track the number of repetitions.

```
int count = 1;  
  
while (count <= 5) {  
    System.out.println(count);  
    count++;  
}
```

Output:

```
1  
2  
3  
4  
5
```

This loop has three important parts:

- Initialization: `int count = 1;`
- Condition: `count <= 5`
- Update: `count++;`

Tracing The Loop

Value of count	Is count <= 5?	Result
1	true	Display 1
2	true	Display 2
3	true	Display 3
4	true	Display 4
5	true	Display 5
6	false	Stop the loop

Infinite Loops

An infinite loop occurs when the condition never becomes false.

```
int count = 1;

while (count <= 5) {
    System.out.println(count);
}
```

The value of count never changes, so the condition `count <= 5` always remains true.

To prevent this problem, update the loop variable inside the loop.

```
count++;
```

A While Loop May Not Execute

Because the condition is checked before the loop body, a while loop may execute zero times.

```
int number = 10;

while (number < 5) {
    System.out.println(number);
}
```

The condition is already false, so the statement inside the loop does not execute.

Let's Practice

Practice 1: Trace The Program

Study the following program:

```
public class LoopTracing {
    public static void main(String[] args) {
        int number = 2;

        while (number <= 8) {
            System.out.println(number);
            number = number + 2;
        }
    }
}
```

Answer the following questions:

1. What is the initial value of number?
2. What condition controls the loop?
3. How does the value of number change?
4. What numbers will the program display?
5. What is the value of number when the loop stops?

Practice 2: Complete The Loop

Complete the missing parts so the program displays the numbers from 1 to 5.

```
public class CompleteLoop {
    public static void main(String[] args) {
        int count = ____;

        while (count ____ 5) {
            System.out.println(count);
            count____;
        }
    }
}
```

Practice 3: Find And Correct The Error

The following program is supposed to display the numbers from 5 down to 1.

```
public class FixTheLoop {
    public static void main(String[] args) {
        int number = 5;

        while (number >= 1) {
            System.out.println(number);
        }
    }
}
```

```
        number++;  
    }  
}  

```

Identify the error and rewrite the incorrect statement.

Practice 4: Create A While Loop

Create a Java program that:

1. Starts with the number 10.
2. Uses a while loop.
3. Displays the even numbers from 10 to 20.
4. Displays one number on each line.
5. Stops after displaying 20.

Resources / References

- [Java Development Kit](#)
- [Visual Studio Code](#)
- [Java while loop documentation](#)
- [Java Scanner class](#)